LINGUISTIC CHARACTERISTICS RELATED TO SUICIDALITY USING AN EXISTING DATA ARCHIVE OF REDDIT POSTS

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Introduction

- Suicide is a widespread public health crisis warranting investigation into its causes and potential interventions.
- Our ability to predict suicide has not improved overall during the past five decades despite much research (Franklin et al., 2017), which has partly encouraged new approaches to studying it.
- As psychological processes are reflected in the language we use (Boyd et al., 2022), we can study suicidality using natural language processing (NLP), which is an area of computer and data science that uses and attempts to "teach" computers to understand human language (IBM, n.d.).
- Social media is a valuable source of naturalistic text data for psychological research that uses natural language processing.
- Previous research has found that there are differences between the language used by suicidal and non-suicidal individuals. The present study explores a sample of approximately two million Reddit posts made by individuals who are and are not suicidal.
- The findings of this line of research have implications for the prediction, detection, and treatment of suicidality.

Method

- The present study analyzes the University of Maryland Reddit Suicidality Dataset, Version 2 (Shing et al., 2018; Zirikly et al., 2019), a dataset of Reddit posts and metadata that contains posts from users that either did post to r/SuicideWatch ("suicidal group") or did not post to r/SuicideWatch or other subreddits related to mental health ("control group").
- We analyzed our data using the Linguistic Inquiry and Word Count, 2022 Version (LIWC-22), a text analysis software that quantifies word use in a given sample of text by calculating the percentage of words in the text that are part of each of its pre-defined dictionary categories, or "variables", relative to the total number of words.
- To investigate differences in language between the suicidal and control groups, we built two logistic regression models that predicted which group a Reddit user was in using either all LIWC-22 variables (Model 1) or a selected subset of ten LIWC-22 variables (Model 2) as predictors.

Differences in the way individuals use language predict whether or not they are suicidal.

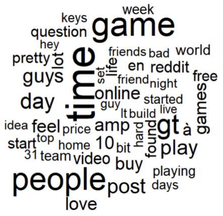

Figure 1

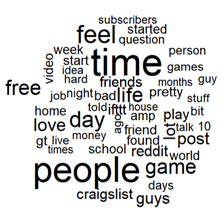

Figure 2

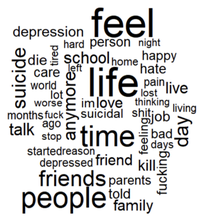

Figure 3

Word clouds of the 50 most frequently used words from:
- all posts by the control group (Figure 1)
- all non-r/SuicideWatch posts by the suicidal group (Figure 2)
- all r/SuicideWatch posts by the suicidal group (Figure 3)

References

1. Boyd, R., Ashokkumar, A., Seraj, S., & Pennebaker, J. (2022). The development and psychometric properties of LIWC-22.
2. Franklin, J. C., Ribeiro, J. D., Fox, K. R., Bentley, K. H., Kleiman, E. M., Huang, X., Musacchio, K. M., Jaroszewski, A. C., Chang, B. P., & Nock, M. K. (2017). Risk factors for suicidal thoughts and behaviors: A meta-analysis of 50 years of research. *Psychological Bulletin*, *143*(2), 187–232.
3. IBM. (n.d.). *What is natural language processing?* Retrieved March 20, 2023, from https://www.ibm.com/topics/natural-language-processing
4. Shing, H.-C., Nair, S., Zirikly, A., Friedenberg, M., Daumé III, H., & Resnik, P. (2018). Expert, crowdsourced, and machine assessment of suicide risk via online postings. *Proceedings of the Fifth Workshop on Computational Linguistics and Clinical Psychology: From Keyboard to Clinic*, 25–36.
5. Zirikly, A., Resnik, P., Uzuner, Ö., & Hollingshead, K. (2019). CLPsych 2019 Shared Task: Predicting the degree of suicide risk in Reddit posts. *Proceedings of the Sixth Workshop on Computational Linguistics and Clinical Psychology*, 24–33.

Additionally, we would like to thank the American Association of Suicidology and the University of Maryland research team responsible for developing and sharing the University of Maryland Reddit Suicidality Dataset, Version 2, for providing us with the data that we used to conduct this research.

Results

- Of the 118 LIWC-22 variables used in Model 1 as predictors of whether a user was in the suicidal or control group, 35 were significant at the $p < 0.05$ level.
- Of the 10 selected LIWC-22 variables used in Model 2 as predictors of whether a user was in the suicidal or control group, 8 were significant at the $p < 0.05$ level.
- Statistics summarizing the predictive abilities of the two models are presented in the following table:

	Model 1	Model 2
accuracy	83.3%	80.5%
sensitivity	84.3%	80.4%
specificity	82.3%	80.7%

* accuracy = proportion of data points correctly classified
* sensitivity = ratio of cases correctly classified as suicidal-group to actual suicidal-group cases
* specificity = ratio of cases correctly classified as control-group to actual control-group cases

Discussion

1. The results of our logistic regression analyses indicate that LIWC-22 variables, and by extension, language use, are able to distinguish between the suicidal and control groups in our data with relatively high accuracy.
2. The LIWC-22 variables that best differentiated between the suicidal and control groups include both content variables (such as "mental", "death", and "Social") and style variables (such as "WC", "negate", and "det"), suggesting that suicidal individuals both talk about different things and talk in different ways compared to non-suicidal individuals.
3. The findings of the present study could be applied to clinical practice by, for example, developing a screening tool for suicidality that analyzes a user's language use and indicates whether they may or may not be at risk for suicidal behavior.